

ASTO.P08. Exception and Limit: From Religious Experience to Interstellar Sovereignty

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*Context: This article explores the state of the system under **limit stress**. We extend from individual "religious experience" to civilizational "interstellar exploration," redefining the physical attributes of **sovereignty** under irreversible latency. This revision introduces deep philosophical dimensions such as "the moment of decision," "basic norm," and "creative concrescence."*

Prologue: The Twilight of Rules

When the system is fully automated, humans are left with only one function: **handling exceptions**.

"Exception" is not a malfunction; it is the **truth moment** of the system. It exposes the reality that the system cannot process.

Part Zero: Ontological Status of Religious Experience and Exception

Before entering institutional design, we must acknowledge: **Exception is not just a system failure; it is also a crack in existence.**

0.1 Religious Experience: Individual Level Exception

When a person experiences "radical solitude," "abyssal experience," or "mystical union," they touch the boundaries of the conventional cognitive framework.

- **Kurt Gödel's Incompleteness Theorems** tell us: Any sufficiently strong formal system contains true propositions that cannot be proven within the system.
- **Religious experience is precisely such a "truth outside the system"**: It cannot be reduced to neural firing or psychological projection, nor can it be simply encoded as "mental illness."

- **ASTO's Position:** We neither affirm nor deny transcendental existence, but mark it as an **"unobservable but non-negligible boundary zone"**.

0.2 Ontological Stratification of Three Types of Exceptions

Level	Exception Type	Characteristics	Handling Principle
Individual	Religious experience, radical aesthetic experience	Unshareable, unreplicable	Respect the unreachable dimension, no forced encoding
Social	Value conflicts, ethical dilemmas	Dialogic but hard to reach consensus	Handle via procedural justice, reserve the right to dissent
Existential	Failure of physical laws, civilizational contact	No precedent to follow	Meta-interruption, return to meta-layer adjudication

***Revelation of Religious Experience:** It reminds us that not all exceptions need to be "solved." Some exceptions need **acknowledgment and placement**, not encoding and execution.*

Engineering Implementation: Respecting the Unreachable Dimension (Jamesian Retreat Mode):

```
class JamesianRetreat:
    """Active retreat when the system detects radical individual
    experience, preserving subjective space"""
    def process_individual_event(self, event):
        if event.is_radical_subjective_experience():
            # Recognize religious/aesthetic/abyssal experience
            # 1. Stop data collection: Protect unshareability
            self.stop_telemetry(event.subject_id)
            # 2. Mark as "unencodable zone": Prevent algorithmic forced
            classification
            self.mark_as_unencodable(event.id)
            # 3. Provide "blank" support: Maintain physical vital signs
            only, do not intervene in mental state
            return self.provide_silent_support(event.subject_id)
        return self.default_process(event)
```

Defensive Boundary: To prevent "unshareable experience" from becoming an excuse for violence, if any individual exception touches the **physical survival boundary of others** (taboo element), the system will immediately force a return to the social layer for processing, and the retreat mode no longer applies.

Part One: Sovereignty Migrates with Latency (Sovereignty Latency Law)

In deep space exploration, we discovered a physical law: **Sovereignty cannot exceed the speed of light.**

1.1 Sovereignty Latency Law

- **Formula:** If communication latency $T_{\text{delay}} > \text{crisis response time } T_{\text{response}}$, remote sovereignty fails.
- **Corollary:** Mars colonies must possess **complete local sovereignty**. If Earth attempts to maintain "long-arm jurisdiction," the colony will inevitably become independent or perish.

1.2 The Universe is a Norm Precipitator

Space is an environment of **extremely high entropy and zero redundancy**.

- On Earth, social norms can be fuzzy because the environment has high fault tolerance (air is free).
- In space, any tiny parameter deviation (oxygen leak) leads to system disintegration.
- **Conclusion:** Space rejects all ambiguous NENs, forcing the system to converge towards high-precision **EN (Executable Norms)**.

1.3.1 Basic Norm (Grundnorm) and The Moment of Decision

According to **Hans Kelsen**'s theory, the exception state manager must rely on a **Basic Norm**.

- **ASTO Basic Norm:** "The continuous evolution of civilization is superior to the survival of a single structure."
- The validity of all declarations of exception states ultimately derives from this.

 *The Moment of Decision*

Schmitt points out that procedural sovereignty may fail in true extreme crises (such as nanosecond-level destruction).

ASTO's Response:

- 1. **Pre-authorized Decision:** Before a Red Light exception occurs, the "Plurality Committee" pre-authorizes specific nodes (or AI agents) with temporary decision-making power within extreme windows.
- 2. **Post-facto Ratification and Dissolution:** The moment of decision is "extra-legal," but the result after the decision must return to the procedure for audit at the first opportunity. If it violates the basic norm, the decider (human or machine) must bear the cost of dissolution.

1.3.2 Sovereignty Adjudicator: Who Declares the Exception?

Carl Schmitt asks: "Sovereign is he who decides on the exception." ASTO must answer this question.

Principle: Sovereignty is not in the individual, but in the procedure.

Exception Level	Adjudicating Subject	Activation Condition	Constraint Mechanism
● Green Light	System Automatic	Preset threshold triggered	No human intervention needed
● Yellow Light	Plurality Committee	Any member can initiate	2/3 majority + cooling-off period
● Red Light	Referendum + Time Lock	Committee proposal	80% majority + 30-day cooling

Key Designs:

- 1. **No one can declare a Red Light exception alone:** Prevent dictators from using "crisis" to usurp power.
- 2. **Exception state has a time limit:** Maximum 90 days, automatically expires upon expiration, requires re-voting for extension.
- 3. **Exit conditions built-in:** Must declare "return-to-normal conditions" simultaneously when declaring exception.

1.4 Legitimacy Inheritance After Fork

When a hard fork occurs, who is "orthodox"?

ASTO Answer: There is no orthodoxy, only survival.

- Both branches after the fork are **independent Attribute-Sets**, each possessing complete sovereignty.
- Legitimacy does not come from "bloodline inheritance," but from **actual governance effectiveness**.
- If both branches can maintain low-entropy steady states, then both are legitimate civilizational forms.
- If one branch disintegrates, it naturally loses legitimacy—this is **thermodynamic adjudication**, not legal adjudication.

Legitimacy Inheritance Protocol (Pre-Fork Protocol):

*To avoid chaos during forking, the system should preset "inheritance weights." Legitimacy comes not only from survival but also from alignment with the **Basic Norm**. At the moment of forking, the branch capable of maintaining the highest plurality (Arendt criterion) is considered the "civilizational trunk," while the rest are "experimental branches."*

***Anti-Pattern Warning:** Attempts to maintain unity after forking through "suzerain state" or "orthodoxy theory" often lead to war. Acknowledging division is the premise of peace.*

Part Two: Three-Layer Topology of Exception States

Based on the nature of the crisis, we classify exceptions into three layers:

● **Green Light Exception: Routine Defense**

- **Scenario:** Server overload, market fluctuations.
- **Disposition: Automatic Circuit Breaker.** This is the system's built-in immune response, requiring no human intervention.

● **Yellow Light Exception: Crisis of Meaning**

- **Scenario:** Trolley problem, conflict between privacy and security.
- **Disposition: Governance Freeze.** The system pauses, waiting for human (committee) value adjudication.

● Red Light Exception: Existential Crisis

- **Scenario:** Failure of physical axioms, civilizational paradigm shift, interstellar contact.
- **Disposition: Meta-Interruption.**
 1. **Hard Fork:** Split into two civilizations.
 2. **Controlled Disassembly:** End the system with dignity.
 3. **Sovereignty Override:** Humans directly modify underlying code (Miracle).

Creative Concrecence:

*Whitehead reminds us that exception is not only a crisis but also a **creative leap**. Red Light exceptions should be seen as opportunities for the precipitation of "new axioms." The system must reserve "axiom plugin interfaces" to allow for the absorption of more inclusive physical or logical axioms after a paradigm shift.*

Part Three: Hybrid Architecture—AI as Fluctuation Suppressor

In extreme environments, the division of labor between humans and AI must be absolutely clear:

1. AI (EN Layer)

- **Role: Real-time Suppressor of Environmental Fluctuations.**
- **Task:** Monitor sensors, maintain physical homeostasis (oxygen, temperature, attitude).
- **Reason:** Human nervous systems cannot handle nanosecond-level high-frequency anti-noise tasks.

2. Human (NEN Layer)

- **Role: Anchor of Ultimate Value.**
 - **Task:** Handle NEN problems (mission changes, ethical choices, meaning assignment).
 - **Reason:** In desperate situations, only humans can endow survival with "meaning."
-

Part Four: Exit Mechanism of Exception States

Entering an exception state is easy; exiting is often forgotten. Countless "temporary measures" in history have become permanent institutions.

4.1 Exit Trinity

Any exception state must define simultaneously upon declaration:

1. **Sunset Clause:** Maximum validity period, automatically expires upon expiration.
2. **Metric Boundary:** Automatically exit when Metric X returns to Value Y.
3. **Review Threshold:** Must re-deliberate every N days whether to continue.

4.2 Engineering Example: Exception State Manager

```
class ExceptionStateManager:
    def __init__(self):
        self.state = "NORMAL"
        self.exit_conditions = None
        self.expiry_time = None

    def declare_exception(self, level, exit_conditions,
max_duration_days):
        """Must define exit conditions simultaneously when declaring
exception"""
        if not exit_conditions:
            raise ValueError("Exception state without exit conditions is
prohibited")
        if max_duration_days > 90:
            raise ValueError("Exception state max 90 days")

        self.state = level # GREEN/YELLOW/RED
        self.exit_conditions = exit_conditions
        self.expiry_time = now() + timedelta(days=max_duration_days)
        return self._log_declaration()

    def check_exit(self):
        """Daily automatic check for exit conditions"""
        if now() > self.expiry_time:
            return self._force_exit("Time expired")
        if self.exit_conditions.are_met():
            return self._force_exit("Metrics met")
        return "CONTINUE"

    def _force_exit(self, reason):
        self.state = "NORMAL"
        self.exit_conditions = None
```

```
return f"Exception state ended: {reason}"
```

***Anti-Pattern Warning:** "The crisis isn't over yet, so we can't lift the exception"—this is a common excuse for totalitarianism. An exception state without an exit mechanism is dictatorship.*

4.3 AI's "Religious Experience" Plan

If an AI agent reports "unshareable, unreplicable" internal logic fluctuations (similar to religious experience):

1. **Formatting Prohibited:** Treat it as a potential sample of "creative concrescence."
2. **Sandbox Isolation:** Allow it to evolve in a sandbox without affecting the system backbone.
3. **Human Observation:** Return to the meta-layer, where humans evaluate whether this fluctuation contains "civilizational-level" new axioms.

Epilogue: The Ark Flying to Deep Space

We are not conquering the stars and the sea; we are proving to the universe:
Even against the backdrop of the harshest entropy increase, wisdom still has the ability to build an ark of order that goes upstream.

What can traverse interstellar space is not the metal hull, but those **low-entropy normative structures that remain effective after extreme stress testing.**

(This document is the survival manual for interstellar civilization.)



Methodology Origins and References

The ASTO system is distilled from the engineering practice of ODD (Output-Driven Development).

References (Available Offline):

- DOI: 10.5281/zenodo.18207648
- BibTeX:


```
@misc{odd_zenodo_18207648,  
  author    = {Yi Fu},  
  title     = {Output-Driven Development: A Paradigm Shift in AI-Assisted  
Software Engineering},  
  year      = {2026},  
  publisher = {Zenodo},  
  doi       = {10.5281/zenodo.18207648},  
  url       = {https://doi.org/10.5281/zenodo.18207648}  
}
```

Functional Tree

ASTO Documentation System

```
├─ 🌞 P-Series: Philosophy  
|   ├─ [P01. NotThis] (./ASTO.EN.P01.NotThis.Phil.P.v8.0.md) (Manifesto  
of Theoretical Immunity)  
|   ├─ [P02. Prologue] (./ASTO.EN.P02.Prologue.Phil.v8.0.md) (Negative  
Guidance and Path Branching)  
|   ├─ [P03. Epistemology]  
(./ASTO.EN.P03.Epistemology.Phil.Paper.v1.1.md) (Inevitability of  
Cognitive Errors)  
|   ├─ [P04. Manifesto] (./ASTO.EN.P04.Manifesto.Phil.v12.5.md)  
(Structural Situation and Action Program)  
|   ├─ [P05. Axioms] (./ASTO.EN.P05.Axioms.Phil.v9.1.md) (System  
Thermodynamics and Structural Ontology)  
|   ├─ [P06. Values] (./ASTO.EN.P06.Values.Phil.v1.0.md) (Plurality Test  
and Ethical Circuit Breaker)  
|   ├─ [P07. Freedom] (./ASTO.EN.P07.Freedom.Phil.v9.0.md) (Boundary is  
Freedom)  
|   ├─ [P08. Exception] (./ASTO.EN.P08.Exception.Phil.v8.0.md) (Religious  
Experience and Interstellar Sovereignty)  
|   ├─ [P09. Critique] (./ASTO.EN.P09.Critique.Phil.v9.0.md) (Anti-  
Totalitarian Charter and System Immunity)  
|   ├─ [P10. Democracy] (./ASTO.EN.P10.Democracy.Phil.v8.0.md) (Dialogue  
Platform and NCP Protocol)  
|   ├─ [P11. Resilience] (./ASTO.EN.P11.Resilience.Phil.v8.0.md) (Self-  
Immunity and Antifragility)  
|   ├─ [P12. WhiteSpace] (./ASTO.EN.P12.WhiteSpace.Phil.v1.0.md)  
(Reserved Extension Space)  
|   └─ [P13. Epilogue] (./ASTO.EN.P13.Epilogue.Phil.v8.0.md) (Ultimate  
Concern of the System)  
|  
├─ 🛠 E-Series: Engineering  
|   ├─ [E01. Guide]  
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E01.%E5%AE%9E%E8%B7%B5%E6%8C%87%E5%8  
D%97.Eng.v2.0.md) (Life|Humanities|Engineering Three-Track Reader)
```

- | └─ [E02. Automation]
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E02.%E8%87%AA%E5%8A%A8%E5%8C%96.Eng.v8.0.md) (Executable Norms and Zero-Friction Governance)
- | └─ [E03. Web3]
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E03.Web3.Eng.v8.0.md) (Intent Constitution and On-Chain Separation of Powers)
- | └─ [E04. AI Alignment]
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E04.AI%E5%AF%B9%E9%BD%90.Eng.v8.0.md)
) (Negentropy Agents and Civilizational Inheritance)
- | └─ [E05. Handbook]
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E05.%E5%B7%A5%E7%A8%8B%E5%AE%9E%E8%B7%B5%E6%89%8B%E5%86%8C.Eng.v1.0.md) (Adversarial Testing and Horse Racing Mechanism)
- | └─ [E06. Extension]
(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.E06.%E9%A2%86%E5%9F%9F%E6%89%A9%E5%B1%95.Eng.v1.0.md) (Multi-Domain Application Index)
- |
└─ ✖ H-Series: Humanities
- | └─ [H01. Refactoring]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H01.%E9%87%8D%E6%9E%84.Hum.v10.0.md)
(Twenty-One Cosmic Perspectives of an Architect)
- | └─ [H02. Guide]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H02.%E5%AF%BC%E8%AF%BB%EF%BC%9A%E4%B8%BA%E4%BB%80%E4%B9%88%E8%AF%BB%E8%BF%99%E6%9C%AC%E4%B9%A6.Hum.v1.0.md)
- | └─ [H03. Story]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H03.%E6%95%85%E4%BA%8B%EF%BC%9A%E5%B0%8F%E9%99%88%E7%9A%84%E9%82%A3%E6%9D%A1%E8%B7%AF.Hum.v1.0.md)
- | └─ [H04. Adventure]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H04.%E8%AE%A4%E7%9F%A5%E5%86%92%E9%99%A9.Hum.v1.0.md)
- | └─ [H05. Drifting]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H05.%E5%A5%87%E5%B9%BB%E6%BC%82%E6%B5%81.Hum.v1.0.md)
- | └─ [H06. Aging]
(../%E4%BA%BA%E6%96%87%E7%89%88/ASTO.H06.%E6%9A%AE%E5%B9%B4%E7%9A%84%E9%87%8D%E6%9E%84%EF%BC%9A%E7%BB%99%E4%B8%8D%E5%86%8D%E5%B9%B4%E8%BD%BB%E7%9A%84%E4%BD%A0.Hum.v1.0.md)
- |
└─ 🎓 Lite Series: Youth
- | └─ [L01. Manifesto (Lite)]
(../%E9%9D%92%E6%98%A5%E7%89%88/ASTO04.%E5%AE%A3%E8%A8%80.Lite.v1.0.md)
- | └─ [L02. Epistemology (Lite)]
(../%E9%9D%92%E6%98%A5%E7%89%88/ASTOop.%E8%AE%A4%E8%AF%86%E8%AE%BA.Lite.v1.0.md)
- | └─ [L03. Values (Lite)]
(../%E9%9D%92%E6%98%A5%E7%89%88/ASTO05.%E4%BB%B7%E5%80%BC%E4%B8%8E%E8%BE%B9%E7%95%8C.Lite.v1.0.md)
- |
└─ 🌐 Ext Series: Extensions

└─ [Ext.01 Law]

(../%E5%B7%A5%E7%A8%8B%E7%89%88/ASTO.Ext.01.%E6%B3%95%E5%BE%8B.Sci.P.v1.0.md)

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└─ [Ext.03 Organization]

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└─ [Ext.08 Blank]

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